

# The ALC\_FACT manual

version 0.1

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## **About the code**

Ada Lovelace Centre: Fitting Atomistic Correlations from Trajectories (ALC\_FACT) is an automatic tool to fit atomistic correlations, combining a robust algorithm with the GNUPLOT software [1]. The fitting parameters are used to extract information of the orientational dynamics and transfer events. Most notably, the code allows computing relevant physical quantities together with uncertainties, as long as the input correlations include error bars.

ALC\_FACT is a serial code written in modern Fortran. Its structure for development and maintenance follows the Continuous Integration (CI) practice and it is integrated within the STFC-GitLab DevOps. The code is open-source subject to license restrictions as appropriate.

## **Disclaimer**

The ALC does not fully guarantee the code is free of errors and assumes no legal responsibility for any incorrect outcome or loss of data.

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# 1 Framework

Molecular Dynamics (MD) simulations with different initial conditions will necessarily sample different regions of the phase space. Consequently, the information derived from the generated trajectories (e.g. time dependent correlations) is expected to differ. This fact is particularly problematic for MD simulations of models whose sizes are compatible with Density Functional Theory (DFT), i.e. sizes of the order of 1000 atoms or less. One possible strategy is executing several MD runs to generate several trajectories, from which the corresponding correlations are computed. These correlations are then used to compute the mean correlation as well as the standard deviation (STD). Depending on the size of the system and the length of the required trajectory, however, this option can be computationally very demanding or even impractical. An alternative is to use a single MD trajectory, divide it in different time segments with the same duration, and perform the data analysis for each of these segments [2]. The computed profiles can then be used to estimate the average and the STD for the correlation under consideration.

Regardless of how averages (and STDs) are obtained, correlations by themselves are not often enough to extract useful information from the simulations. In fact, one also needs to fit the data using a functional form that complies with a given physical model, so physical quantities can be derived from the fitted parameters. ALC\_FACT offers an automatic tool for such a purpose.

## 2 Description of settings

In this section we provide a description of the directives for execution. These directives must be defined within the SETTINGS file, located in the folder from which the code is executed. Although the settings for fitting and analysis depends on the type of correlation, the order of the directives within the file is irrelevant. Moreover, the code is flexible enough to recognize directives independently of capitalization. For example, the **filename** directive can be defined as **filename**, **FILENAME**, **fileName**, etc.

In the following, we first detail the minimum set of directives to analyse the average correlations. Then, we describe how the input correlation errors are automatically included in the analysis. For completeness, and to facilitate their identification, we also report the implemented directives in Table 1.

### 2.1 Fitting and data analysis of average correlations

Firstly, the user must define the **what\_to\_fit** directive to specify the quantity under consideration. The implemented options are the following:

- **ocf**: Orientational Correlation Function for reactive (**reactive\_species .True.**) [3] and non-reactive species (**reactive\_species .False.**) [4, 5].
- **tcf**: Transfer Correlation Function (TCF) [6].
- **spcf**: Special Pair Correlation Function (SPCF) [6].
- **msd**: Mean Square Displacement (MSD).

As described, the analysis of Orientational Correlation Functions (OCFs) requires the specification of the **reactive\_species** directive. When setting **reactive\_species .True.**, the special pair (formed between the reactive species and the closest non-reactive species) is the quantity of relevance, related the orientation of the reactive species [3]. When setting **reactive\_species .False.**, in contrast, it is the orientation of the non-reactive species the quantity under consideration [4, 5, 7]. In other words, reactive and non-reactive OCFs correspond to different physical quantities and require different analyses. Both types of OCFs can be computed from MD trajectories and their profiles can be compared with orientational anisotropies obtained from 2D-IR experiments.

On the other hand, TCFs, SPCFs and MSDs are purely computational derived quantities: while TCFs and SPCFs necessarily refer to reactive systems, MSDs are only computed for non-reactive species (independently if the whole system is reactive or not). For this reason, specification of the **reactive\_species** directive is not needed for these three quantities. Finally, if a MSD analysis is requested, the code will ask for the **&msd** block, which must include the information of the related set of MSD coordinates and the units (see below).

The code also requires the definition of the **species\_name** directive, which must be a string used to name the species under consideration. The name of the file with the input data for analysis must be defined with the **filename** directive. This file must be located in the same folder than the SETTINGS file.

Together with the definition of the **what\_to\_fit**, the user must define the **fitting\_function** directive. The implemented options for **fitting\_function** for OCF offer the use of 1 and 2 decay exponentials (**1exp** and **2exp**, respectively) that tend to zero or a constant (**to\_zero** and **to\_constant** respectively). This makes a total of four different type of functions that involve exponentials (see Table 2). For example, if the user decides to fit the data using 1 exponential that tends to a constant, the option for **fitting\_function** must be set to **1exp\_to\_constant**.

The same fitting options, except **2exp\_to\_constant**, are available for TCFs and SPCFs. The only available choice for MSDs is **linear**. We refer to Table 2 for details.

The format of the input data (within the filename specified via the **filename** directive) must be given in columns. To indicate which column correspond to the recorded time and average correlation values, the user must set the **time\_column\_number** and **avg\_column\_number** directives with the corresponding column number in the input file. It is user's responsibility to define these directives correctly.

Finally, the definition of the **max\_iterations** directive (maximum number of iterations for fitting) is enough to fit and analyse the average correlation. Consequently, the fitted parameters and derived physical quantities will be printed to the OUTPUT file. The raw data and the fitted values are printed by default to the RAW\_DATA and FITTED\_DATA files, respectively.

## 2.2 Including correlation errors in the fitting

In addition to the directives described in the previous section, the inclusion of the correlation errors in the fitting requires the definition of extra settings. Although they have not the same meaning, error and standard deviation (STD) are referred interchangeably hereinafter. The user must first define the **std\_column\_number** directive indicating the column number of the input file that contains the error for each correlation point. In ALC\_FACT, the inclusion of errors in the fitting is conducted by considering multiple profiles, taking  $N_{sp}-1$  deviations (i.e. samples;  $N_{sp}$  is an odd number) from the average correlation  $C_{avg}(t)$ . Each deviation  $C_j(t)$  is given by the following expression:

$$C_j(t) = C_{avg}(t) + \frac{[2(j-1) - (N_{sp}-1)]}{(N_{sp}-1)} \omega S(t) \quad j = 1, \dots, N_{sp} \quad (1)$$

where  $S(t)$  is the input standard deviation and  $\omega$  a real number, which indicates how much of the  $S(t)$  is considered. With such construction, we have  $C_k(t) = C_{avg}(t)$  for  $k = \frac{N_{sp}-1}{2} + 1$ . The values of  $N_{sp}$  and  $\omega$  must be defined with the **std\_samples** and **max\_times\_std** directives, respectively. The values for these two directives will depend on the problem and correlation under consideration, but it is recommended not to set  $\omega$  larger than 1. If there are no problems with the input data nor mistakes with the settings of **what\_to\_fit** or **fitting\_function** directives, ALC\_FACT will fit each of the  $C_j(t)$  profiles, from which the computed average and STD of fitted parameters (and physical quantities) will be printed to the OUTPUT file. It is important to mention that if **std\_samples** is set to 1, the correlation errors will not be considered and the analysis will be executed only for the average values.

## 2.3 Extra directives

By default, the fitting analysis uses the data corresponding to the whole time range provided in the input file. Nevertheless, depending on the case, it might be convenient to fit and analyse the input data by constraining the time domain. In ALC\_FACT this is possible with the **start\_time** and **end\_time** directives. If **start\_time** is not defined, the data from the lowest recorded time will be considered. Likewise, if **end\_time** is not defined, all data up to the largest recorded time will be considered. Details of the selected time domain will be printed to the OUTPUT file.

Relevant to computed OCF data, it might be convenient to include superfast response times associated with librations. This is possible by setting the **fit\_superfast\_time** directive to **.True.**, which adds an extra exponential to the fitting function, as shown in Table 2. To achieve convergence in the fittings, it is required to include raw data for very short times. If problems are found, the code will inform the user accordingly by printing warnings in the OUTPUT file.

ALC\_FACT has a robust internal initialization of variables depending on settings for the **what\_to\_fit** or **fitting\_function** directives. Nevertheless, in the exceptional case that the fittings do not converge, the user can initialise the involved parameters via the **&input\_parameters** block.

The parameters derived from each fitting are printed to the OUTPUT file in a table format, together with the Sum of Square of Residuals (SSR) divided by the Number of Degrees of Freedom (NDF), which provides an estimate of the quality of the fitting. Optionally, the user can set the **print\_average\_ssr** to **.True.** to print the average SSR/NDF.

Finally, the user can prevent the printing of the RAW\_DATA and FITTED\_DATA files by setting the **plot\_RAW\_data** and **plot\_fittings** to **.False.** (both are **.True.** by default).

Table 1: Description of implemented directives for the SETTINGS file.

| Directive/Block           | Format         | Description and Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>what_to_fit</b>        | <b>String</b>  | Type of correlation to be fitted and analysed. Implemented options: <ul style="list-style-type: none"> <li>• <b>ocf</b>: Orientational Correlation Function (OCF).</li> <li>• <b>tcf</b>: Transfer Correlation Function (TCF).</li> <li>• <b>spcf</b>: Special pair Correlation Function (SPCF).</li> <li>• <b>msd</b>: Mean Square Displacement (MSD; needs the <b>&amp;msd</b> block).</li> </ul>                                                                                                                                                                                                                                                                                                                                          |
| <b>reactive_species</b>   | <b>Logical</b> | Only relevant for OCF analysis. Specifies if the OCFs corresponds to reactive ( <b>.True.</b> ) or non-reactive ( <b>.False.</b> ) species.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>species_name</b>       | <b>String</b>  | Label of the atomistic/molecular species under consideration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>filename</b>           | <b>String</b>  | Name of the file that contains the input data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>fitting_function</b>   | <b>String</b>  | Functional form to fit the data. Implemented options (see Table 2): <p><b>Valid for OCF, TCF, and SPCF</b></p> <ul style="list-style-type: none"> <li>• <b>1exp_to_zero</b>: one exponential term, the function tends to zero.</li> <li>• <b>1exp_to_constant</b>: one exponential term, the function tends to a constant.</li> <li>• <b>2exp_to_zero</b>: two exponential terms, the function tends to zero.</li> </ul> <p><b>Only valid for OCF</b></p> <ul style="list-style-type: none"> <li>• <b>2exp_to_constant</b>: two exponential terms, the function tends to a constant.</li> </ul> <p><b>Only valid for MSD</b></p> <ul style="list-style-type: none"> <li>• <b>linear</b>: MSD is set to depend linearly with time.</li> </ul> |
| <b>time_column_number</b> | <b>Integer</b> | Column number of the input file corresponding to the reported time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                                                                                                                                                                                                                                                                                                                                              |            |                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>avg_column_number</b>                                                                                                                                                                                                                                                                                                                     | Integer    | Column number of the input file corresponding to the average values of the correlation.                                                        |
| <b>std_column_number</b>                                                                                                                                                                                                                                                                                                                     | Integer    | Column number of the input file corresponding to the STD for the correlation. If <b>std_samples</b> is equal to 1, this directive is neglected |
| <b>max_iterations</b>                                                                                                                                                                                                                                                                                                                        | Integer    | Maximum number of iterations to converge the fitting.                                                                                          |
| <b>std_samples</b>                                                                                                                                                                                                                                                                                                                           | Integer    | Number of profiles (samples) generated from the average correlation and STDs. Value of $N_{sp}$ in Eq. (1).                                    |
| <b>max_times_std</b>                                                                                                                                                                                                                                                                                                                         | Real       | Maximum times the STD values. This is $\omega$ in Eq. (1).                                                                                     |
| <b>start_times</b>                                                                                                                                                                                                                                                                                                                           | Real Units | Set the lower time bound for fitting data. Valid Units are ps and fs. If not defined, the lowest recorded time is considered.                  |
| <b>end_times</b>                                                                                                                                                                                                                                                                                                                             | Real Units | Set the upper time bound for fitting data. Valid Units are ps and fs. If not defined, the largest recorded time is considered.                 |
| <b>fit_superfast_time</b>                                                                                                                                                                                                                                                                                                                    | Logical    | Includes an addition exponential decay for very short time response.                                                                           |
| <b>plot_raw_data</b>                                                                                                                                                                                                                                                                                                                         | Logical    | Controls the printing of the RAW_DATA file (.True. by default)                                                                                 |
| <b>plot_fittings</b>                                                                                                                                                                                                                                                                                                                         | Logical    | Controls the printing of the FITTED_DATA file (.True. by default)                                                                              |
| <b>print_ssr_average</b>                                                                                                                                                                                                                                                                                                                     | Logical    | Prints the average SSR/NDF to the OUTPUT file (only for converged fittings).                                                                   |
| <b>&amp;msd</b> Block needed in case the <b>what_to_fit</b> is set to <b>msd</b> .<br><b>select</b> String      # MSD coordinates: Options: <b>x</b> , <b>y</b> , <b>z</b> , <b>xy</b> , <b>xz</b> , <b>yz</b> and <b>xyz</b><br><b>units</b> String      # MSD Units: Options: <b>Angstrom^2</b> and <b>Bohr^2</b> .<br><b>&amp;end_msd</b> |            |                                                                                                                                                |
| <b>&amp;input_parameters</b> Block to initialise parameter.<br># Only in the rare case when the fittings fail to converge. What<br># parameters $A_i$ and $T_i$ are needed depend on the<br># correlation and functional form. See the Table 2.<br><b>&amp;end_input_parameters</b>                                                          |            |                                                                                                                                                |

### 3 Description of fitting formulae and physical models

Table 2 reports the fitting formulae implemented in ALC\_FACT for the different options of the **what\_to\_fit** and **fitting\_function** directives. The parameters  $A_i$  and the times  $T_i$  are defined internally unless the user set the **&input\_parameters** block. Only for OCFs, super fast times can be included in the fitted with the **fit\_superfast\_time** directive.

Once the correlation is fitted, the resulting parameters are of limited use unless they become input values to physical models. Such models allow extracting relevant physical quantities that can support the interpretation of fundamental processes at the atomistic level. In the following sections we briefly describe the physical models adopted in ALC\_FACT to analyse the different correlations.

Table 2: Fitting formulae for the different options of *what\_to\_fit* and *fitting\_function*. The formulae denoted with the symbol (\*) correspond to the case when the *fit\_superfast\_time* directive is set to *True*.

| <i>what_to_fit</i>       | <i>fitting_function</i> | Formulae<br>Parameters to fit: A0, A1, A2, A3, T1, T2, T3                                                                                               |
|--------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>ocf</i>               | <i>1exp_to_zero</i>     | $f(t)=A1*\exp(-t/T2)$<br>$f(t)=A1*\exp(-x/T2)+(1-A1)*\exp(-x/T1-x/T2)$ (*)                                                                              |
|                          | <i>1exp_to_constant</i> | $f(t)=A2+(A1-A2)*\exp(-t/T2)$<br>$f(t)=A2+(A1-A2)*\exp(-t/T2)+(1-A1)*\exp(-t/T1-t/T2)$ (*)                                                              |
|                          | <i>2exp_to_zero</i>     | $f(t)=A2*\exp(-t/T3)+(A1-A2)*\exp(-t/T2-t/T3)$<br>$f(t)=A2*\exp(-t/T3)+(A1-A2)*\exp(-t/T2-t/T3)+$<br>$(1-A1)*\exp(-t/T1-t/T2-t/T3)$ (*)                 |
|                          | <i>2exp_to_constant</i> | $f(t)=A3+(A2-A3)*\exp(-t/T3)+(A1-A2)*\exp(-t/T2-t/T3)$<br>$f(t)=A3+(A2-A3)*\exp(-t/T3)+(A1-A2)*\exp(-t/T2-t/T3)+$<br>$(1-A1)*\exp(-t/T1-t/T2-t/T3)$ (*) |
| <i>tcf</i> & <i>spcf</i> | <i>1exp_to_zero</i>     | $f(t)=\exp(-A1*t)$                                                                                                                                      |
|                          | <i>1exp_to_constant</i> | $f(t)=A0*\exp(-A1*t)+(1-A0)$                                                                                                                            |
|                          | <i>2exp_to_zero</i>     | $f(t)=A0*\exp(-A1*t)+(1-A0)*\exp(-A2*t)$                                                                                                                |
| <i>msd</i>               | <i>linear</i>           | $f(t)=A0+A1*t$                                                                                                                                          |

### 3.1 Orientation of non-reactive species: the wobbling-in-a-cone model

ALC\_FACT uses the wobbling-in-a-cone model to analyse OCFs of non-reactive species. This model is extensively used to quantify the orientational dynamics through relaxation times and order-parameters. Such order parameter can be related to a cone angle within which the main axis of the (molecular) species diffuses and wobbles. Details of formulation and relevant physical variables can be found in Ref. [4, 5]. Table 3 summarises the computed physical quantities for different options of *fitting\_function*.

Table 3: Computed physical quantities from the wobbling-in-a-cone model for the analysis of OCFs. The possible options of *fitting\_function* are considered separately.

| <i>fitting_function</i>                                                                                          | Computed quantities (see below)                              |
|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <i>1exp_to_zero</i>                                                                                              | theta_lib, Dw, Tau_cone [theta_cone=90 degrees in this case] |
| <i>1exp_to_constant</i>                                                                                          | theta_lib, theta_cone, Dw, Tau_cone                          |
| <i>2exp_to_zero</i>                                                                                              | theta_lib, theta_cone, Dw, Tau_cone, Tau_host                |
| <i>2exp_to_constant</i>                                                                                          | theta_lib, theta_cone, Dw, Tau_cone, Tau_host                |
| (*) For all options, Tau_lib is also computed if the <i>fit_superfast_time</i> directive is set to <i>True</i> . |                                                              |

theta\_lib: librational angle (degrees)

theta\_cone: cone angle (degrees)

Dw: Diffusion associated with the final cone angle wobbling (1/ps)

Tau\_cone: Effective relaxation time for the wobbling cone (ps)

Tau\_host: Effective relaxation time for the host (ps)

Tau\_lib: Superfast librational time (ps)

### 3.2 Orientation of reactive species: relaxation times

In contrast to the analysis of OCFs for non-reactive species, there is no an equivalent wobbling-in-a-cone model when reactive species are involved. In fact, the orientation of reactive species is determined by the orientation of the special pair, whose dynamics cannot be described in terms of a diffusive motion within a cone [3]. Consequently, the analysis can only extract relaxation times, which are described in Table 4 for different options of *fitting\_function*.

Table 4: Computed quantities from the fitting of OCFs for reactive species. The possible options of *fitting\_function* are considered separately.

| <i>fitting_function</i>                                                                                       | Computed quantities (see below) |
|---------------------------------------------------------------------------------------------------------------|---------------------------------|
| <i>1exp_to_zero</i>                                                                                           | Tau_net                         |
| <i>1exp_to_constant</i>                                                                                       | Tau_net, A_inf                  |
| <i>2exp_to_zero</i>                                                                                           | Tau_fast, Tau_slow              |
| <i>2exp_to_constant</i>                                                                                       | Tau_fast, Tau_slow, A_inf       |
| (*) For all options, Tau* is also computed if the <i>fit_superfast_time</i> directive is set to <i>True</i> . |                                 |

Tau\_net: Net relaxation time (ps)

A\_inf: Remanent orientational fraction (dimensionless)

Tau\_fast: Fast relaxation time (ps)

Tau\_slow: Slow relaxation time (ps)

Tau\*: Superfast relaxation time (ps)

### 3.3 Special pair and transfer correlations: first-order kinetics approximation

In ALC\_FACT, the parameters from the fittings of SPCFs and TCFs are used together with the first-order kinetic approximation [6] to derive transfer rates and characteristic times. Table 5 summarises the computed physical quantities for different options of *fitting\_function*.

Table 5: Computed transfer rates and characteristic times from the fitting of SPCFs and TCFs. Different choices of *fitting\_function* are considered.

| <i>fitting_function</i> | Computed quantities (see below)    |
|-------------------------|------------------------------------|
| <i>1exp_to_zero</i>     | kf, 1/(kf)                         |
| <i>1exp_to_constant</i> | kf, 1/(kf), kb, 1/(kb)             |
| <i>2exp_to_zero</i>     | kf, 1/(kf), kb, 1/(kb), k2, 1/(k2) |

kf: Forward rate constant (in 1/ps)

kb: Backward rate constant (in 1/ps)

k2: Second forward rate constant (in 1/ps)

1/(kf): Forward characteristic time (in ps)

1/(kb): Backward characteristic time (in ps)

1/(k2): second forward characteristic time (in ps)

### 3.4 Mean Square Displacement: diffusion coefficient of non-reactive species

From the linear fitting of the MSD (the only valid option for MSDs), the rate A1 (see Table 2) is used to compute the diffusion coefficient,  $D$ , via the Einstein's relation [8] as follows

$$D = \frac{A1}{2N_{\text{dim}}} \quad (2)$$



where  $N_{\text{dim}}$  refers to the dimensionality for which MSD was computed. The value of  $N_{\text{dim}}$  is derived from the the setting of the **select** directive inside the **&msd** block: equal to 1 (if **select** is set to **x**, **y** or **z**), 2 (if **select** is set to **xy**, **xz** or **yz**) and 3 (if **select** is set to **xyz**). The valid units for D are **Angstrom^2/ps**.

### 3.5 Computing average and errors

The last sections describe how physical quantities are derived from the fitted parameters. This is already sufficient if one only consider the average correlation values. However, the analysis differs if the STD for the input correlation is included. In this section, we briefly describe how ALC\_FACT automatically processes the input STD to derive average values and errors only for reference purposes. In fact, the user does not need to worry about these internal operations.

As shown in sec. 2.2, the STD of the input correlations is used to built  $N_{\text{sp}}$  profiles that depart from the average correlation  $C_{\text{avg}}(t)$ , according to Eq. (1). Once each correlation  $C_j(t)$  is fitted, the resulting parameters  $\{A_i^{(j)}\}$  and  $\{T_i^{(j)}\}$  are obtained from the settings of the **what\_to\_fit** and **fitting\_function** directives (see Table 2). To derive average values and STD for the fitted parameters it is important to remark that the weight of each profile  $C_j(t)$  is not the same: the larger the departure of profile  $C_j(t)$  from  $C_{\text{avg}}(t)$  the lower its weight. Assuming a normal-like distribution, we define the weights for each profile,  $w^{(j)}$ , with the following expression

$$w^{(j)} = \exp\left[-\frac{\omega^2 f_{(j)}^2}{2}\right] \quad \text{where} \quad f_{(j)} = \frac{[2(j-1) - (N_{\text{sp}} - 1)]}{(N_{\text{sp}} - 1)}. \quad (3)$$

Thus,  $w < 1$  for all  $C_j(t)$  expect for the average correlation ( $w=1$ ). The average value and STD for each fitted parameter of Table 2 is computed using these weights. Taking A1 as example, the average value  $\langle A1 \rangle$  is given by

$$\langle A1 \rangle = \frac{\sum_{j=1}^{N_{\text{sp}}} w^{(j)} A1^{(j)}}{\sum_{j=1}^{N_{\text{sp}}} w^{(j)}} \quad (4)$$

while its STD by

$$A1(STD) = \sqrt{\frac{\sum_{j=1}^{N_{\text{sp}}} w^{(j)} (A1^{(j)} - \langle A1 \rangle)^2}{\sum_{j=1}^{N_{\text{sp}}} w^{(j)}}}. \quad (5)$$

The same weights and formulae is applied to calculate the average and STD for each physical quantity, using the physical quantities obtained separately from each profile  $C_j(t)$ .

### 3.6 Extra considerations

The design of ALC\_FACT focused on providing a robust automatic tool. Nevertheless, several problems could manifest when executing the fitting: errors in the input data, wrong or inadequate settings for the **what\_to\_fit** and **fitting\_function** directives, lack of convergence, etc. The code will print warning and/or error messages, which are meant to instruct the user if problems are encountered. Likewise, the code also offers a robust evaluation of the fitted parameters as well as the derived physical quantities, which are compared to check their validity. In case of problems, the code will print messages to guide the user wit the definition of the directives.

The development of ALC\_FACT also focused on providing a robust infrastructure to guide the user with the definition of directives. That is, the user could well start with a blank SETTINGS file and run the code, which will obviously abort, but print an error message with the required directives to be defined. The code will keep aborting until the minimum number of directives are set. Likewise, the code will abort and (hopefully) instruct the user if a directive was defined incorrectly. The order of the required directives do not need to follow a particular order as long as they are all defined.

In the **examples** folder the user will find several examples of the different quantities that can be fitted.



## 4 References

- [1] Thomas Williams, Colin Kelley, and many others. Gnuplot 4.4: an interactive plotting program. <http://gnuplot.sourceforge.net/>, March 2010.
- [2] Seonmyeong Kim, Xiangwen Wang, Jeongmin Jang, Kihoon Eom, Simon L. Clegg, Gun-Sik Park, and Devis Di Tommaso. Hydrogen-bond structure and low-frequency dynamics of electrolyte solutions: Hydration numbers from ab initio water reorientation dynamics and dielectric relaxation spectroscopy. *ChemPhysChem*, 21(20):2334–2346, 2020.
- [3] Paul B. Calio, Chenghan Li, and Gregory A. Voth. Resolving the structural debate for the hydrated excess proton in water. *Journal of the American Chemical Society*, 143(44):18672–18683, 2021. PMID: 34723507.
- [4] K. Kinosita, S. Kawato, and A. Ikegami. A theory of fluorescence polarization decay in membranes. *Biophysical Journal*, 20(3):289–305, 1977.
- [5] G. Lipari and A. Szabo. Effect of librational motion on fluorescence depolarization and nuclear magnetic resonance relaxation in macromolecules and membranes. *Biophysical Journal*, 30(3):489–506, 1980.
- [6] Joseph A. Morrone, Kiryn E. Haslinger, and Mark E. Tuckerman. Ab initio molecular dynamics simulation of the structure and proton transport dynamics of methanol-water solutions. *The Journal of Physical Chemistry B*, 110(8):3712–3720, 2006. PMID: 16494428.
- [7] I-Chun Lin, Ari P. Seitsonen, Ivano Tavernelli, and Ursula Rothlisberger. Structure and dynamics of liquid water from ab initio molecular dynamics—comparison of blyp, pbe, and revpbe density functionals with and without van der waals corrections. *Journal of Chemical Theory and Computation*, 8(10):3902–3910, 2012. PMID: 26593030.
- [8] Mark E. Tuckerman. *Statistical Mechanics: Theory and Molecular Simulation*. Oxford University Press, Oxford, first edition, 2010.